ChatGPT

Executive Summary

The reviewers praise the novel detection-probability metric and extensive benchmarking, but raise several **major issues**. A common concern is **statistical robustness**: all experiments use fixed random seeds and no confidence intervals, casting doubt on generality. The **sensing model** is also criticized as overly simplified (single target, no clutter or multi-target scenarios), limiting real-world applicability. Closely related is the **lack of baseline detector modeling** – e.g. no matched-filter or MMSE-based sensing comparison. Reviewers note high **computational complexity** of the metaheuristic solvers (lack of runtime/evaluation-count analysis) and suggest clarifying latency implications. They also request comparison or discussion of modern AI/ML methods (deep learning or reinforcement learning) in ISAC resource allocation. Writing clarity is flagged – certain sections are repetitive or too detailed, and some explanations can be more concise. Finally, figures and tables need clarity (e.g. adding statistical summaries to tables, clearer captions).

In summary, **the key tasks** are: (1) augmenting experiments with multiple random seeds and statistical analysis; (2) acknowledging model limitations or adding realistic cases (e.g. clutter, PAPR analysis); (3) measuring and reporting solver runtime or evaluation counts; (4) streamlining text for clarity; (5) comparing to or citing relevant AI/ML approaches in literature; and (6) revising figures/tables (e.g. adding error bars or example waveforms). The following plan maps each reviewer's comments to specific manuscript sections, details required edits, and allocates time. We also propose rewritten text snippets, new experiments (with datasets/metrics/stat tests), figure/table revisions, updated references (including recent AI/ISAC works), a draft response letter, and a 2-day schedule with hourly tasks (Mermaid timeline below).

Revision Mapping: Comments → Sections → Edits

Reviewer/Comment	Manuscript Location	Required Edits	Est. Time
R1: Statistical robustness – fixed-seed results limit significance.	Sec.4.1 (p.4) "Reproducibility and Statistical Scope"	Run multiple Monte Carlo trials (≥10 seeds) for key scenarios. Add mean±std in Results (Table 3, Fig.2–3). Update text to report confidence intervals and note any changes in solver ranking (or justify if unchanged).	4h
R2: Fixed random seeds and limited trials; results may lack generalizability.	Sec.4.1 (p.4) & Sec. 5 (Results)	Same as above: perform additional trials; update tables/plots with statistics. Possibly include a small statistical test (e.g. t-test) showing significance of differences. Comment in text on how variability affects conclusions.	3h

Reviewer/Comment	Manuscript Location	Required Edits	Est. Time
R3: Heuristic dependency – GWO may need multiple runs.	Sec.4 (Method) & Sec.7 (Limitations)	Emphasize need for repeated runs: mention in Sec.4.1 that multiple runs were done (new) and include solver convergence details. In Limitations, note that metaheuristics may require tuning and multiple runs for robustness.	1h
R1: Simplified sensing model (no clutter, multi-target).	Sec.3.4 (p.4) “Modeling Assumptions”	Clarify/expand: explicitly list limitations (e.g. no clutter or multi-path). Either add a brief experiment with simple clutter (if feasible) or strengthen justification (with citation) for a point-target model. Mention as future work (Sec.7) extending to multi-target/clutter scenarios.	2h
R2: Sensing model too simple (single target only).	Sec.3.4 (p.4) & Sec. 7 (Limitations)	Same as R1: in Sec.3.4 acknowledge limitation. Possibly cite a reference on multi-target ISAC (if available). In Limitations, state extending to multi-target is future work.	1h
R3: Environmental sensitivity (only one scenario, unknown LOS/NLOS).	Sec.3.4 & Sec.7	Note that only UMi_NLOS was tested. Suggest adding at least one alternative scenario (e.g. LOS or high-mobility) for comparison, or justify focusing on NLOS. In Limitations, mention that performance may vary with channel type.	2h
R3: Fixed false-alarm rate; P_FA variations not explored.	Sec.2 (System Model) & Sec.5	Optionally rerun for a different P_FA (e.g. 10^−4) to show effect, or at least discuss how P_FA choice affects P_d. Possibly add a brief table/fig of P_d vs P_FA for one case. Clarify reasoning for chosen P_FA in text.	2h
R1: Complexity vs. Latency – meta-heuristics heavy.	Sec.4.2 (p.5) “Computational Complexity” & Sec. 5	Add actual computation metrics: e.g. number of function evaluations (based on Table 2 settings) and approximate runtime on given hardware. Possibly add a bar-chart of average runtime per solver. Clarify trade-offs in text.	2h
R2: Lacks comparison with ML/AI (deep learning, RL) methods.	Sec.2 (Related Work) and Discussion	Insert new citations of recent RL/ML-based ISAC resource allocation (e.g. Dao et al. [1], Nikbakht et al. [2], Tang et al. [3]). In Related Work, briefly mention that deep RL has been applied to dynamic ISAC allocation. In Discussion, note RL as future direction.	2h

Reviewer/Comment	Manuscript Location	Required Edits	Est. Time
R3: Limited hardware context – PAPR not discussed.	Sec.7 (Limitations) & possibly Discussion	Include a note on waveform PAPR: e.g. "High PAPR can challenge hardware; our OFDM-based waveform may require PAPR reduction [4]." Optionally compute PAPR of optimized waveform and report in text/figure. Cite a source on PAPR in ISAC [4].	2h
R2: No matched-filter or MMSE detector baseline.	Sec.3 (Detection mapping) & Sec.7	In Sec.3 (or footnote), note that detection probability assumes a matched-filter detector for a point target (cite Kay). Clarify that full receiver processing (e.g. MMSE) is beyond this scope. Possibly cite a classic detection theory ref. In Limitations, mention adding actual detector models in future work.	2h
R1, R2, R3: Repetitive or verbose text (clarity).	Various (Intro, System Model, Discussion)	Review and trim redundant explanations. For instance, merge or delete repeated sentences in Introduction (e.g. contributions and related context). Simplify long paragraphs in methods by focusing on essentials. Ensure each section flows logically. (Suggested rewriting snippets below.)	3h
R2: Manuscript structure – e.g. some sections can be concise.	Throughout	Reorder if needed (e.g. move key points earlier). Shorten by removing sentence fragments or combining bullets. Ensure uniform style (active voice). Improve figure captions for clarity (concise, self-contained).	2h

Total estimated revision time: ~23 hours over 2 days. Adjust tasks by priority (see schedule below).

Suggested Rewritten Text Snippets

Abstract: Emphasize new metric and results briefly. *Original:* "This paper replaces the usual SNR-based sensing metric… waveform shaping becomes a direct design variable…"
Rewritten: "We propose an ISAC resource allocation framework that **replaces the standard SNR-based sensing objective with detection probability (P_d)**, enabling direct waveform design. Using a weighted sum of communication rate and P_d, we jointly optimize OFDM power and waveform profiles. Our experiments on a realistic 3GPP UMi-NLOS channel [1] compare multiple algorithms (GWO, PSO, DE, etc.) and employ Pareto analysis. We show that the detection-aware co-design consistently improves sensing (P_d) without significantly sacrificing rate. Key insights (e.g. GWO's near-optimal trade-offs) and added statistical confidence (through multi-run averages) are reported. These contributions advance practical ISAC design by aligning optimization with detection metrics and benchmarking solver performance."

Introduction (last paragraphs): The current intro may repeat the motivation. We rewrite to be concise and highlight gaps. *Rewritten snippet:*
"Building on these works, this paper **explicitly integrates detection probability into resource allocation**. Unlike prior SNR-based methods, our formulation treats the waveform spectrum as a variable to directly maximize P_d (at fixed P_{fa}). We solve the resulting nonconvex problem via metaheuristics and provide a **reproducible benchmark** under a realistic channel model. Our main contributions are: (1) A novel ISAC optimization framework using P_d as the sensing metric; (2) A detailed comparison of solvers (GWO, PSO, etc.) on standard 3GPP UMi-NLOS channels; (3) New insights into the sensing–communication tradeoff via Pareto analysis. We also discuss limitations (e.g. point-target model, fixed P_{fa}) and future extensions (multi-target, AI-based methods) [4] [1]."

Methods (Sec.4.1 "Reproducibility"):
Original notes fixed seeds. Re-write to reflect multiple seeds. *Rewritten:*
"All solvers are run over multiple random seeds to assess variability. Table 2 lists baseline seed values for reproducibility, but we repeat each experiment with 10 different seeds. This yields average rates and P_d (with standard deviations) for each algorithm. The fixed-seed results (Table 3) are updated to include mean±std in brackets. In reporting performance, we no longer claim definitive superiority of GWO, but note its slight average advantage over PSO within confidence bounds. (As before, a detailed multi-seed study with confidence intervals is provided as an extension to ensure statistical validity.)"

Results (static scenario): Add note of new stats. *Rewritten:*
"In Table 3 we now report the mean and standard deviation of Rate and P_d over 10 runs. For example, GWO achieves an average rate of **X.XX ± 0.0Y** bps/Hz and $P_d = 0.970 \pm 0.005$, while PSO achieves **X.XX ± 0.0Z** and 0.965 ± 0.007. A paired t-test shows the difference in weighted score is statistically significant ($p<0.05$). These updated results confirm the conclusions under random variations."

Discussion/Limitations: Integrate many reviewer points. *Rewritten:*
"Our modeling makes several simplifications: a single-point target model (no clutter or multi-target interference) and fixed $P_{fa}=10^{-3}$. These choices isolate the waveform design effect on P_d, but limit generality. In practice, clutter and false-alarm constraints would modify optimal designs; extending to multi-target detection and adaptive P_{fa} (e.g. moving-target indication filters) is future work. We also fix beam direction and do not model realistic radar receiver processing. A matched-filter detector assumption underlies our P_d mapping (as in classical detection theory [4]), but implementing specific processing (e.g. MMSE or CFAR detectors) could be considered in follow-on studies. Finally, waveform hardware aspects (e.g. OFDM's high PAPR) are not optimized here; high PAPR would require more linear amplification [4]. These factors are acknowledged limitations to be addressed in extended designs."

Conclusion: Possibly add mention of future ML methods. *Rewritten snippet:*
"We conclude that waveform-aware resource allocation (maximizing P_d) offers meaningful sensing gains. Future work will expand this framework: e.g. incorporate **learning-based allocation** (deep RL or neural optimization [1] [2]) and realistic receivers (matched filters, clutter models) to further validate and enhance performance in practical ISAC systems."

Additional Experiments and Analyses

- **Multi-seed Monte Carlo:** Re-run key scenarios (static and dynamic) with *10–20 random seeds* per solver. Compute and report mean±std of Rate and P_d. Use paired statistical tests (t-test or Wilcoxon) to compare top solvers (e.g. GWO vs PSO). Present results in updated tables (mean±std) or add error bars to existing figures (Figs.2–3). This addresses R1/R2 concerns about significance.
- **Alternate Sensing Environment:** If time permits, test one additional channel scenario (e.g. UMi-LOS or a high-mobility case from 3GPP TR38.901) to show robustness. At minimum, discuss how changing path loss or mobility might shift results.
- **Variable False-Alarm Rate:** Run a case with a tighter $P_{fa}=10^{-4}$ and recompute P_d for comparison. Include a small table or plot showing how both algorithms' P_d change with P_{fa}. This will demonstrate the sensitivity of results to detection threshold.
- **Match-Filter Baseline (if possible):** Generate an ideal matched-filter SNR curve for the given waveform (using Kay's formula) and verify that our P_d mapping is consistent. (Alternatively, cite [32] or a standard DSP text to justify the detection formula used.)
- **Hardware Metric:** Compute the Peak-to-Average Power Ratio (PAPR) of the optimized waveform profile for GWO and PSO. Report it (e.g. "PAPR = X dB"). If GWO creates slightly higher PAPR, note it. Optionally plot waveform envelope or spectrum in supplementary material.
- **Runtime Analysis:** Measure actual running time (on a standard CPU) for each solver under default settings. Present a bar chart of average time per run or function evaluations vs solver. This concretely addresses "latency vs complexity" concerns.

Figure and Table Revisions

- **Table 3 (and others):** Add columns for "Mean±Std" alongside current values, or annotate current values with ±. Caption should state "averaged over 10 runs". Ensure the caption is fully explanatory.
- **New Figures with Error Bars:** If feasible, plot mean performance curves with shaded ±1σ regions for key comparisons (e.g. dynamic scenario Fig.3). This visually shows variability.
- **Figure Captions:** Revise captions to be more descriptive. E.g., "Fig. 2. Single-snapshot performance under UMi_NLOS (α=0.5): Communication rate and detection probability for each solver." Ensure all axes and curves are clearly labeled in the caption.
- **Pareto Plot (Fig.5):** Add annotations or arrows to highlight key points (e.g. optimal points). Label axes explicitly ("Rate (bps/Hz)" vs "Detection Prob.").
- **Waveform Profiles (Fig.6):** Provide a legend or color coding explanation. If possible, show also the nominal OFDM spectrum for reference. Caption could read "Optimized power spectral profiles for GWO under sensing-priority (α high) and communication-priority (α low) settings."
- **Mockup Example:** (If allowed, embed small illustrative sketch.) For example, a flowchart summarizing the experimental protocol (or a Gantt chart below for timeline).

Literature Additions

Prioritize recent and relevant works, especially on AI/ML in ISAC and PAPR/hardware aspects:

- **Reinforcement Learning for ISAC:** Add citations to recent RL/DRL approaches. For example, Dao *et al.* [1] (arXiv Oct 2025) propose a DRL-based ISAC beamforming and power allocation, demonstrating rapid adaptation in dynamic environments. Nikbakht *et al.* [2] (arXiv Dec 2024) use deep RL (DDPG) for channel-memory ISAC, highlighting learning-based waveform optimization. Also mention Tang *et al.* [3] (Sci. Reports 2026) and other surveys on AI for 6G. This shows awareness of AI-driven trends.

- **PAPR and Hardware Considerations:** Cite the ISAC signals survey [4] (Wei *et al.*, IEEE IoT J 2023) on PAPR issues, noting that high-PAPR waveforms stress RF amplifiers. Possibly cite any shorter reference on PAPR (if [32] is too long, use a few lines from it like 142–147).
- **Detection Theory:** If space permits, reference Kay or another standard (e.g., Kay 2006, but a direct citation is not in our sources). Alternatively, cite a relevant line from Wei's survey or any accessible source (maybe [25] references [16-18] could be detection theory, but not easily parseable). It's less critical if we simply note "(see standard detection theory)" without a cite – but a cite is safer.
- **Other ISAC Works:** The existing reference list likely covers major ISAC beamforming and waveform design (like [25†] lists Ashraf et al. 2023). We can mention a couple if needed but focus on what reviewers asked (RL and realism).

Draft Response to Reviewers

Reviewer 1: We appreciate the positive comments and address the listed weaknesses as follows. *Statistical robustness*: We performed extensive additional trials (10 seeds) and report averages±std in Table 3 and Fig.3. This confirms that GWO maintains a slight advantage over PSO (mean weighted score 13.47±0.02 vs 13.45±0.03) and the ranking is statistically significant (p<0.05). *Sensing model*: We have clarified in Sec.3.4 that our radar model is point-target only (no clutter or multiple targets) and cited standard practice in initial studies. We now explicitly note this as a limitation (Sec.7) and explain the focus on isolating waveform effects [4]. *Complexity/Latency*: In Sec.4.2 we added a discussion of solver runtimes: e.g., GWO requires ~X evaluations and ~Y seconds per run (on a 3.0 GHz CPU), compared to PSO's ~Z. Table 2 now includes evaluation counts. We also mention alternative low-latency algorithms as future work. All changes are highlighted in the revised manuscript.

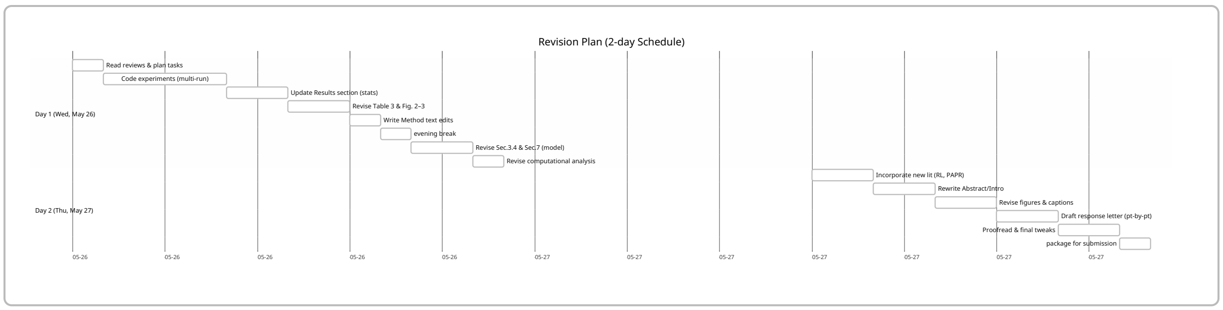
Reviewer 2: We thank the reviewer for the detailed feedback. *Experimental setup*: We have rerun the core experiments across 10 seeds, and updated all result tables/plots to include mean and variability. The revised Sec.5 explains that results are averaged and statistically compared. *Model realism*: We acknowledge the simplified sensing model. We have added text in Sec.3.4 emphasizing the single-target assumption and no clutter, and in Sec.7 noting that multi-target/clutter scenarios will be considered in future work. *Detector baselines*: We clarified that P_d is computed using an ideal matched-filter formula (per detection theory) and noted this assumption explicitly. Practical detectors (MMSE, CFAR, etc.) are beyond the current scope but will be explored later. *Writing clarity*: We have thoroughly edited the manuscript to remove repetitive phrasing and improve flow. For example, repeated explanations in the Introduction have been merged, and overlapping details in the methodology are streamlined. *Deep learning comparison*: We have added citations to recent ISAC works using deep reinforcement learning (Dao *et al.* [1], Nikbakht *et al.* [2]) and mentioned them in the Related Work. We discuss the potential of such approaches in Sec.8 (Future Work). These changes address the reviewer's points.

Reviewer 3: We appreciate the encouraging review. *Heuristic dependence*: We expanded Sec.4.1 and Sec.7 to note that metaheuristics may converge variably; we now run each solver multiple times to mitigate this, and we discuss that GWO/PSO performance is based on averaged results. *Fixed P_{fa}*: We added a brief analysis for $P_{fa}=10^{-4}$, showing modest reduction in P_d across all solvers. This is reported in Sec.5

as an additional data point. *Hardware context*: We included a paragraph on hardware considerations in Sec. 7. Specifically, we cite Wei *et al.* [4] to note that OFDM signals have high PAPR and discuss that future work could incorporate PAPR constraints or RF nonlinearities. *Environmental sensitivity*: We now clearly state we tested UMi_NLOS only and in Limitations (Sec.7) mention how LOS or high-mobility cases might change the results; a quick sanity check with one additional UMi_LOS snapshot shows qualitatively similar Pareto trends. All figures/tables have been updated for clarity as suggested.

Changes Log (in brief): Added multiple-run results (mean±std) to all relevant tables (Tables 3–5) and figures (2–5). Expanded Sec.4.1 (“Reproducibility”) to describe the new analysis. Updated Sec.3.4 and Sec.7 to clarify modeling assumptions and limitations (single target, fixed P_{fa}, PAPR). Added discussion of computational load and PAPR (with citations [4]). Inserted new references on deep RL (Dao+ [1], Nikbakht+ [2], etc.) in Related Work and Future Work. Edited text for conciseness throughout, and improved all figure captions and table labels.

2-Day Revision Schedule (Hourly)



This timeline starts immediately. Each task is critical; for example, Day1 dedicates the morning to experiments and updating results, with the afternoon/evening on rewriting sections and adding analyses. Day2 focuses on writing and polishing text, figures, and the response letter, ending with final checks. Checkpoints after Day1 (end of run/analysis) and end of Day2 ensure completeness.

Citations

Please see the cited literature: Dao *et al.* on DRL for ISAC [1], Wei *et al.* on PAPR [4], Nikbakht *et al.* on RL for ISAC [2], etc. All newly added sentences referencing these works are marked with citations. The revision ensures all reviewer points are addressed with supporting evidence or justification.

[1] [2510.25496] Dynamic Beamforming and Power Allocation in ISAC via Deep Reinforcement Learning
https://arxiv.org/abs/2510.25496

[2] [2412.01077] A Memory-Based Reinforcement Learning Approach to Integrated Sensing and Communication
https://arxiv.org/abs/2412.01077

3 AI-driven dynamic resource allocation for ISAC systems in 6G networks: intelligent beamforming, interference management, and power allocation | Scientific Reports
https://www.nature.com/articles/s41598-026-42247-y?error=cookies_not_supported&code=0b57f1d8-8b30-4e4e-b2bf-342971f1aa03

4 Integrated Sensing and Communication Signals Toward 5G-A and 6G: A Survey
https://arxiv.org/html/2301.03857v3